

■ Ocean Hazard Reporting & Social Media Analytics

Day WSIE Project Tracker Report

Field	Details
Batch	III AIML_A – 2028
Tracking Period	June 01, 2025 – June 13, 2025
Project Duration	June 01 – June 30, 2025
Project Type	Solo Project

Social Media Analytics

Machine Learning

Real-Time Monitoring

Hazard Detection

DAY
01

■ Planned Activity

Project Title Finalization

Jun-01 | Jun-01

■ Expected Deliverable

Approved Project Title

What I Did

On Day 1, I brainstormed and finalized the project title. I reviewed multiple AI and social media analytics ideas, evaluated technical feasibility, and selected a project that addresses a real-world coastal safety challenge using modern technologies.

Activities Breakdown

- Brainstormed 10+ project ideas focused on disaster tech and AI-driven analytics
- Evaluated each idea on feasibility, innovation, and academic value
- Shortlisted 3 topics: Smart Traffic AI, Hospital AI Portal, Ocean Hazard Analytics Platform
- Finalized: 'Ocean Hazard Reporting & Social Media Analytics'
- Got faculty approval for the title

Why This Topic?

Reason	Explanation
Real-World Impact	Coastal hazards (tsunamis, rip currents, oil spills) threaten lives; AI + social media can enable faster early warnings
Technology Combination	Combines NLP + ML models + GIS dashboards + real-time feeds — novel and technically challenging
Industry Relevance	Applicable to coastal authorities, disaster agencies, researchers, and coastal communities
Academic Rigor	Covers all required modules: DB, Frontend, Backend, AI, NLP, Security

DAY
02

■ Planned Activity

Requirement Gathering

Jun-02 | Jun-02

■ Expected Deliverable

Problem Statement

Problem Statement

Problem Statement: Coastal regions worldwide are highly vulnerable to ocean hazards such as tsunamis, storm surges, coastal flooding, rip currents, oil spills, and marine pollution. Traditional monitoring systems depend on official sensors and periodic manual reports, which often fail to capture localized incidents in real time. At the same time, social media platforms carry eyewitness reports during hazard events, but this data is unstructured, noisy, and impossible to process manually at scale. There is a pressing need for an intelligent, automated platform capable of collecting social media signals, classifying hazards using AI, and delivering real-time situational awareness to authorities and coastal communities.

Stakeholder Analysis

Stakeholder	Requirements
Coastal Communities	Real-time hazard alerts, evacuation signals, reliable safety updates
Coastal Authorities	Dashboard for monitoring hazard reports, incident logs, historical data
Researchers	AI-predicted hazard trends, sentiment data, ocean risk assessment insights
System Administrator	Manage data sources, user accounts, AI model configs, system health

Functional Requirements

- User registration, login, and role-based profile management
- Real-time social media data collection and ingestion pipeline
- AI-based hazard classification (tsunami, flooding, rip current, oil spill, etc.)
- Sentiment and panic-level analysis from social media posts
- Fake news / misinformation detection for hazard reports
- Automated alert and notification system for high-risk detections
- GIS-based hazard visualization on interactive maps
- Historical trend analysis and reporting module

Non-Functional Requirements

Requirement	Specification
System Uptime	99.9% availability — critical safety system
Alert Response Time	< 3 seconds from detection to notification
Data Security	End-to-end encryption + role-based JWT auth
Scalability	Support 10,000+ social media posts/min and 100+ concurrent users

DAY
03

■ Planned Activity

Objective Definition

Jun-03 | Jun-03

■ Expected Deliverable

Project Objectives

Project Objectives

Obj #	Objective Description
OBJ-01	Collect and ingest hazard-related social media posts in real time using APIs and scrapers
OBJ-02	Preprocess and store structured and unstructured data in a hybrid database (PostgreSQL + MongoDB)
OBJ-03	Detect and classify ocean hazards from social media text using BERT-based NLP models
OBJ-04	Perform sentiment and panic-level analysis on public responses during hazard events
OBJ-05	Identify and filter misinformation and fake hazard reports using AI-based detection
OBJ-06	Visualize hazard locations and severity on interactive GIS maps in real time
OBJ-07	Generate automated alerts and notifications for high-risk detected events
OBJ-08	Provide predictive risk assessment and historical trend analysis for coastal regions

SMART Goals

Criterion	Description
Specific	Build an ocean hazard monitoring platform using social media AI analytics in 30 days
Measurable	All 8 objectives completed with working NLP pipeline and live dashboard demo
Achievable	Technologies are well-documented; I have prior Python/React/FastAPI skills
Relevant	Directly addresses coastal safety gaps using state-of-the-art AI and NLP
Time-Bound	Complete by June 30, 2025 with full submission and demo

DAY

04

■ Planned Activity

User & Module Identification

Jun-04 | Jun-04

■ Expected Deliverable

Module List

User Roles

User Role	Description	Access Level
General Public / Coastal Resident	Views hazard alerts and safety status for their area	Basic
Coastal Authority Officer	Monitors dashboard, validates reports, responds to incidents	Medium
Researcher / Analyst	Analyzes hazard trends, accesses sentiment and risk data	Medium
System Administrator	Manages data sources, users, AI model configs, full system	Full

Module List

Module #	Module Name	Description
M-01	User Management	Registration, Login, JWT, Role-based access control
M-02	Data Collection Layer	Ingests social media posts via Twitter/X API, Reddit, and scrapers
M-03	Data Processing Layer	Cleans, tokenizes, and preprocesses text data for NLP models
M-04	AI / NLP Analytics	BERT-based hazard classification, sentiment analysis, fake news detection
M-05	Geolocation Extraction	Extracts location entities from posts and maps them to coordinates
M-06	Alert & Notification	Triggers alerts and sends notifications on high-risk detections
M-07	Dashboard & Visualization	Interactive GIS maps, hazard feeds, sentiment charts, analytics
M-08	Reporting Module	Generates reports for auditing, export, and trend analysis

DAY

05

■ Planned Activity

Use Case Diagram Preparation

Jun-05 | Jun-05

■ Expected Deliverable

CRUD APIs

Use Case Summary

Actor	Use Case
Public User	Login / View Hazard Map / Subscribe to Alerts / View Reports
Authority Officer	Login to Dashboard / Monitor Live Hazard Feed / Validate Reports / Respond to Incidents
Authority Officer	Generate Incident and Risk Assessment Reports
Researcher / Analyst	Analyze Historical Hazard Data / View AI Predictions / Export Data
Admin	Manage Users and Roles / Configure Data Sources / Update AI Model Parameters
System (AI)	Auto-detect hazards from social media feeds / Classify hazard types
System (Alerts)	Auto-generate warnings and send notifications to relevant stakeholders

CRUD API Matrix

Entity	Create	Read	Update	Delete
User	POST /api/users/register	GET /api/users/{id}	PUT /api/users/{id}	DELETE /api/users/{id}
Hazard Post	POST /api/posts	GET /api/posts/{id}	PUT /api/posts/{id}	DELETE /api/posts/{id}
Hazard Detection	POST /api/detections	GET /api/detections/{id}	PUT /api/detections/{id}	—
Alert	POST /api/alerts	GET /api/alerts	PUT /api/alerts/{id}	—
Hazard Location	POST /api/locations	GET /api/locations/{id}	PUT /api/locations/{id}	DELETE /api/locations/{id}
Report	POST /api/reports	GET /api/reports/{id}	—	—
Sentiment Result	POST /api/sentiment	GET /api/sentiment/{id}	—	—

DAY
06

■ Planned Activity

Database Requirement Analysis

Jun-06 | Jun-06

■ Expected Deliverable

Table List

Database Tables Identified

Table Name	Purpose	Key Columns
users	All user accounts and roles	user_id, name, email, role, password_hash
social_posts	Raw collected social media posts	post_id, platform, content, author, posted_at, location_text
hazard_detections	AI-classified hazard results per post	detection_id, post_id (FK), hazard_type, confidence, detected_at
sentiment_results	Sentiment and panic scores per post	sentiment_id, post_id (FK), sentiment, panic_score, analyzed_at
fake_news_flags	Misinformation detection results	flag_id, post_id (FK), verdict, confidence, flagged_at
hazard_locations	Geolocated hazard incident areas	location_id, hazard_type, coordinates, risk_level, reported_at
alerts	System-generated hazard alerts	alert_id, hazard_type, risk_level, message, sent_at, acknowledged
reports	Exported safety and audit reports	report_id, generated_by, type, file_path, created_at
system_logs	System activity and error logs	log_id, action, user_id, timestamp, details

Database Design Decisions

- PostgreSQL: Used for structured relational data (users, detections, alerts, reports)
- MongoDB: Used for raw unstructured social media posts and AI prediction results
- Redis: Used for caching, real-time notifications, and session management
- All PostgreSQL tables follow 3NF (Third Normal Form) for minimal redundancy
- Auto-increment integers used as primary keys for simplicity and performance
- Timestamps stored in UTC format for consistency across time zones
- Indexed columns: post_id, detected_at, hazard_type, risk_level for fast query performance

DAY

07

■ Planned Activity

ER Diagram Design

Jun-07 | Jun-07

■ Expected Deliverable

ER Diagram

Entity-Relationship Diagram

The ER diagram captures all entities and their relationships. Core entities and attributes are listed below.

■ USERS	
Attribute	Type / Description
user_id (PK)	INT, NOT NULL, AUTO_INCREMENT
name	VARCHAR(100), NOT NULL
email	VARCHAR(150), UNIQUE, NOT NULL
password_hash	TEXT, NOT NULL
role	ENUM (public, officer, researcher, admin)
created_at	TIMESTAMP, DEFAULT NOW()
is_active	BOOLEAN, DEFAULT TRUE

■ One USER generates many ALERTS (1:N) | One USER generates many REPORTS (1:N)

■ SOCIAL_POSTS	
Attribute	Type / Description
post_id (PK)	INT, NOT NULL, AUTO_INCREMENT
platform	ENUM (twitter, reddit, instagram, facebook)
content	TEXT, NOT NULL
author_handle	VARCHAR(100)
location_text	VARCHAR(200) — raw location string from post
posted_at	TIMESTAMP, NOT NULL

■ One SOCIAL_POST produces one HAZARD_DETECTION (1:1) | One SOCIAL_POST has one SENTIMENT_RESULT (1:1)

■ HAZARD_DETECTIONS	
Attribute	Type / Description
detection_id (PK)	INT, NOT NULL, AUTO_INCREMENT
post_id (FK)	INT → social_posts
hazard_type	ENUM (tsunami, coastal_flooding, storm_surge, rip_current, oil_spill, pollution)
confidence_score	FLOAT — 0.0 to 1.0
detected_at	TIMESTAMP

■ One HAZARD_DETECTION may trigger one ALERT (1:0..1)

DAY
08

■ Planned Activity

Database Schema Creation

Jun-08 | Jun-08

■ Expected Deliverable

SQL Schema

SQL Schema Overview

The PostgreSQL schema was created with all tables, constraints, indexes, and foreign key relationships as defined below.

Schema: Relationships Map

From Table	To Table	Relationship	Key
users	reports	1:N — user generates reports	user_id → generated_by
users	alerts	1:N — user receives alerts	user_id → recipient_id
social_posts	hazard_detections	1:1 — post has one detection	post_id → post_id
social_posts	sentiment_results	1:1 — post has one sentiment result	post_id → post_id
social_posts	fake_news_flags	1:0..1 — post may be flagged	post_id → post_id
hazard_detections	hazard_locations	1:0..1 — detection maps to location	detection_id → detection_id
hazard_detections	alerts	1:0..1 — detection triggers alert	detection_id → detection_id

Index Strategy

Table	Index Column(s)	Reason
users	email	Fast login lookup
social_posts	platform, posted_at	Time-series queries and platform filtering
hazard_detections	post_id, hazard_type, detected_at	Dashboard filtering by type and time
alerts	risk_level, sent_at	Real-time alert sorting
hazard_locations	risk_level, coordinates	GIS map fast queries
sentiment_results	post_id, sentiment	Sentiment trend analysis

DAY

09

■ Planned Activity

UI Wireframe Design

Jun-09 | Jun-09

■ Expected Deliverable

Page Layouts

Wireframe Page List

Page #	Page Name	User Role	Key Elements
P-01	Login Page	All	Email/password form, role selector, login button
P-02	Main Dashboard	Officer/Researcher	Live hazard feed, risk map, active alert feed, stats
P-03	Hazard Monitoring Page	All	Hazard list, AI confidence, severity badges, filters
P-04	Interactive Map Page	Officer/Researcher	GIS map with hazard pins, heatmap, severity color-coding
P-05	Sentiment Analysis Page	Researcher	Sentiment charts, panic index, word clouds, timeline
P-06	Alert Center	Officer/Admin	Active alerts list, acknowledge button, severity filter
P-07	Historical Reports Page	Researcher/Admin	Date-range picker, trend charts, hazard history, export
P-08	Social Media Analytics	Researcher/Admin	Platform stats, post volume, hashtag trends, geo heatmap
P-09	Admin Panel	Admin	User management, AI config, data source settings, logs

UI Design Principles

- Responsive design for desktop and tablet (control rooms and field operations)
- Color coding: Red = High risk / tsunامي, Orange = Storm surge, Yellow = Medium, Green = Low, Blue = Info
- Dashboard-centric UI: primary view is the live GIS hazard map and real-time post feed
- Real-time updates via WebSocket connection without page refresh
- React.js with Tailwind CSS for rapid, consistent UI development
- Dark mode support for 24/7 monitoring environments

DAY
10

■ Planned Activity

Login & Dashboard UI Design

Jun-10 | Jun-10

■ Expected Deliverable

UI Screens

Login Screen Design

Component	Description
Logo & Title	App logo, 'Ocean Hazard Reporting System' header in dark navy with wave icon
Email Input	Placeholder: 'Enter your email', format validation
Password Input	Masked input with show/hide toggle
Role Selector	Dropdown: Public User / Authority Officer / Researcher / Admin
Login Button	Primary ocean-blue button, disabled until form is valid
Forgot Password	Link below button, redirects to email reset flow
Error Messages	Red inline validation messages for invalid inputs

Main Dashboard UI Components

Component	Description
Header Bar	App title, user avatar, notification bell (with count), logout
Hazard Map (Main)	Interactive GIS map with coastal zones color-coded by hazard risk level
Stats Cards	Cards showing: Active Hazards, Posts Analyzed Today, Alerts Sent, System Status
Live Hazard Feed	Scrollable list of recent AI-detected hazard posts with severity badges
Sentiment Chart	Real-time donut/bar chart showing positive/negative/panic sentiment split
Hazard Type Breakdown	Bar chart of hazard types detected in the last 24 hours

Authority Officer Dashboard — Additional Components

- Live GIS map with hazard pins color-coded by type and severity
- Incident response panel: sortable by severity, time, and hazard category
- Statistics bar: Total posts analyzed, Active hazard alerts, High-risk detections today
- Alert notification popup: Auto-appears for High-risk detection events

DAY

11

■ Planned Activity

Navigation & Form Design

Jun-11 | Jun-11

■ Expected Deliverable

UI Prototype

Navigation Structure

Navigation Item	Public User	Officer/Researcher	Admin
Home / Dashboard	■	■	■
Hazard Monitor	■	■	■
Interactive Map	■	■	■
Sentiment Analysis	View only	Full Access	Full Access
Alert Center	View only	Full Access	Full Access
Historical Reports	■	■	■
Social Media Analytics	■	View only	Full CRUD
Admin Panel	■	■	■
Profile / Settings	■	■	■

Key Forms Designed

Form Name	Fields	Validation
User Registration	Name, Email, Phone, Password, Role	Email format, phone 10 digits, password strength
Alert Acknowledgement	Alert ID (auto), Action Taken, Notes, Officer ID (auto-fill)	Notes required for High-risk alerts
Report Generator	Report Type, Date Range, Platform Filter, Format (PDF/CSV)	Date range must be valid, type required
AI Config Form	Model Type, Confidence Threshold (0.5–0.99), Data Sources	Threshold must be 0.5–0.99 range
Hazard Submission	Hazard Type, Location, Description, Evidence Image	All required fields, valid coordinates

Prototype Flow

- Figma prototype: All 9 screens connected with click-through interactions
- Public User flow: Login → Dashboard → View Hazard Map → View Alert Details → Subscribe
- Officer flow: Login → Dashboard → View Hazard Alert → Validate Report → Acknowledge & Report
- Admin flow: Login → Admin Panel → Manage Data Sources → Manage Users → Generate Report

DAY
12

■ Planned Activity

Design Review

Jun-12 | Jun-12

■ Expected Deliverable

Design Approval

Design Review Checklist

Review Area	Status	Comments
Project Title	■ Approved	Title is clear, specific, and socially impactful
Problem Statement	■ Approved	Well-defined scope with real-world coastal safety significance
Objectives (8)	■ Approved	All SMART objectives met and clearly defined
Module List (8)	■ Approved	All modules accounted for and described
Use Cases	■ Approved	Covers all actors and interactions comprehensively
CRUD APIs	■ Approved	REST conventions followed correctly
Database Tables (9)	■ Approved	3NF normalized for SQL; NoSQL for unstructured data
ER Diagram	■ Approved	All entity relationships correctly mapped
SQL Schema	■ Approved	Constraints and foreign keys properly defined
UI Wireframes	■ Approved	9 screens fully designed for all user roles
UI Screens	■ Approved	Login and Dashboard components complete
UI Prototype	■ Minor Fix	Sentiment chart animation needs refinement on mobile
Overall Design	■ Approved	Ready to proceed to development phase

Faculty Review Feedback

1. Strong project with clear real-world coastal safety applicability. Social media integration is well-thought-out.
2. Ensure AI module includes fallback logic for low-confidence predictions to avoid false positives triggering unnecessary alerts.
3. Alert system must have escalation levels: local notification → authority officer → emergency services.
4. Overall: Design phase is well-executed and technically sound. Approved for development phase.

DAY

13

■ Planned Activity

Frontend Environment Setup

Jun-13 | Jun-13

■ Expected Deliverable

React Project Setup

Frontend Environment Setup

Tool / Package	Version	Purpose
Node.js	v20.x LTS	JavaScript runtime environment
npm	v10.x	Package manager
React.js	v18.2	Frontend framework
Vite	v5.x	Build tool (faster than CRA)
Tailwind CSS	v3.4	Utility-first CSS framework
React Router DOM	v6.x	Client-side routing and navigation
Axios	v1.6	HTTP client for REST API calls
Recharts / Chart.js	v3.x	Sentiment and hazard data visualization
Leaflet / React-Leaflet	v4.x	Interactive hazard location maps
Mapbox GL JS	v3.x	Advanced geospatial GIS visualization
Socket.io-client	v4.x	Real-time WebSocket hazard feed updates
React Toastify	v10.x	Alert and notification toasts
ESLint + Prettier	Latest	Code quality and formatting

Project Folder Structure

```
ocean-hazard-frontend/
├── src/
│   ├── components/ # Reusable UI components (HazardCard, SeverityBadge, AlertPopup)
│   ├── pages/ # Route-level pages (Dashboard, HazardMonitor, SentimentAnalysis)
│   ├── services/ # Axios API call functions (hazardService, alertService)
│   ├── context/ # React Context (AuthContext, AlertContext)
│   ├── hooks/ # Custom hooks (useHazardFeed, useSentiment)
│   ├── utils/ # Helper functions, validators, risk calculators
│   └── App.jsx # Root component with routes
├── public/ # Static assets (favicon, index.html)
└── package.json # Dependencies manifest
```

Environment Variables (.env)

- VITE_API_BASE_URL=http://localhost:5000/api
- VITE_MAPS_API_KEY=[Mapbox / OpenLayers API Key]
- VITE_WS_URL=ws://localhost:5000/ws (WebSocket for real-time hazard feed)
- VITE_ML_SERVICE_URL=http://localhost:8000 (Python AI/NLP microservice)
- VITE_SOCIAL_API_KEY=[Twitter/X Bearer Token for social media ingestion]

Days 1–13 Summary

Day	Date	Planned Activity	Expected Deliverable
01	Jun-01	Project Title Finalization	Approved Project Title
02	Jun-02	Requirement Gathering	Problem Statement
03	Jun-03	Objective Definition	Project Objectives
04	Jun-04	User & Module Identification	Module List
05	Jun-05	Use Case Diagram Preparation	CRUD APIs
06	Jun-06	Database Requirement Analysis	Table List
07	Jun-07	ER Diagram Design	ER Diagram
08	Jun-08	Database Schema Creation	SQL Schema
09	Jun-09	UI Wireframe Design	Page Layouts
10	Jun-10	Login & Dashboard UI Design	UI Screens
11	Jun-11	Navigation & Form Design	UI Prototype
12	Jun-12	Design Review	Design Approval
13	Jun-13	Frontend Environment Setup	React Project Setup